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Review

Deepfake video detection methods, approaches, and challenges

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ABSTRACT

Deepfake technology creates highly realistic manipulated videos using deep learning models, which makes distinguishing between authentic and fake content extremely difficult. This technology can negatively affect society by breaching privacy and spreading misinformation. This paper presents a comprehensive survey of the recent deepfake video detection approaches and methods. Each deepfake video method is analyzed according to its ability to generalize diverse deepfake fabrication techniques and real-world scenes. We reviewed around 103 articles which eventually shrunk down to 73 based on the screening criteria like abstract/title/irrelevant focus/duplication. The study primarily covers audio-based, visual-based, and multi-modal detection methods. Also, it discusses the usage of Convolutional Neural Networks (CNNs), frequency-domain analysis, and audio-visual synchronization in deepfake video detection and evaluates the strengths and shortcomings of these techniques. Moreover, the study explores major issues such as low resolution, video compression, and adversarial attacks, which prove to be a barrier to making deepfake video detection processes robust. By connecting findings from numerous studies, this research draws attention to the development of standard benchmarking SOPs and multi-modal detection techniques to improve detection performance.

1. Introduction

Deepfake technologies can create highly realistic manipulated videos using deep learning models, which makes distinguishing between authentic and fake content extremely difficult. The integrity of internet information and security systems is jeopardized by alterations to faces and voices [1,2]. Believable audio or visual content is being created through deep learning including voice replication, text-to-speech conversion, voice transformation, sound editing, language conversion, and segmented Deepfakes [3,4]. The production of complex deepfakes by non-experts is facilitated by the rise of affordable technology and accessible editing apps, which exacerbates the challenge faced by forensic tools and experts. Despite significant progress being made in the detection of deepfakes, many detection systems are still facing challenges with adaptability, immediate detection, and resistance to malicious attacks due to their reactive nature [5]. The widespread availability of affordable and advanced mobile devices, such as touchscreen phones, is being coupled with the easy accessibility of face and audio editing apps like FaceApp [6], as well as sophisticated neural network frameworks such as dynamic identity recognition networks [7] and vocoder-based voice synthesis technology (Vocoder-Synth) [8]. This has enabled complex deepfakes and digitally altered faces to be created even by amateurs. Significant challenges are being posed to current investigative technologies and forensic experts by this proliferation. The ongoing evolution of deepfakes is indicating a rise in advanced fake news, where disinformation is being spread by various actors, including automated accounts, external actors, partisan media, conspiracy advocates, and online agitators through the use of realistic Deepfakes [9,10].

Detecting deepfakes is considered essential for preventing the misapplication of computational intelligence in video manipulation [11]. Although attempts to detect deepfake videos have been applied such as the utilization of graph representation and graph neural networks for deepfake detection [12], there are still notable challenges. According to [13], current deepfake detection have achieved about 65 % accuracy in real-world scenarios as compared to near-perfect accuracy in a controlled setting. A variety of methods have been suggested to identify and expose deepfake videos but these procreative models evolve at such a pace that it has become toilsome to keep updating the precision of deepfake detection tools [15]. Initially, detection models possessed limited capacity to identify certain artificially crafted content pieces [14]. Over time, deepfake generation technology escalated to the level that the content rarely appeared contrived entailing the need for advanced detection techniques to combat diverse content-altering methods [16]. Crafting precise detection tools is a challenge considering how accurately deepfake tools can alter and customize facial features, speech, and expressions [17].

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Deepfake video detection can be classified into three categories in terms of approach: Feature-Based Methods (Handcrafted Features) [18], Deep Learning-Based Methods (Model-Based) [19], and Hybrid Methods (Combining Feature-Based and Deep Learning Approaches) [20]. Manipulation algorithms identify artificially constructed features (Feature-based approach) [18], while the model-based approach relies on learning patterns procured from huge datasets of original and fake videos to differentiate between the actual and fabricated content [19]. The third approach comprises Hybrid methods [20], which is the amalgamation of both features-based and model-based approaches. This method utilizes the strengths of both first two methods combined rendering precise and improved detection. Deepfake video detection methods are tested in a controlled environment but the real-world scenario is often different, therefore there is a contrast between the detection done in a controlled environment and the real world. Low-quality, compressed videos circulated on social media further intricate deepfake video detection [21]. Furthermore, low-resolution deepfake videos are often created from multiple low-quality resources therefore conventional detection tools cannot recognize which parts are original and fabricated [22]. The latest research has accentuated the significance of crafting advanced detection techniques that are strong enough to generalize numerous types of deepfake content and low-quality compressed videos [23].

This paper introduces a systematic evaluation of recent advancements in deepfake detection techniques, with a specific focus on their ability to address real-world challenges such as generalization, adaptability, and resistance to adversarial attacks. Unlike prior studies, this paper provides a comprehensive analysis of the evolving landscape of deepfake detection by categorizing methods into visual, audio, and hybrid approaches while highlighting their applicability across diverse datasets and uncontrolled environments. By emphasizing the practical implications of these advancements, the paper underscores the critical need to refine detection techniques to mitigate the growing risks posed by sophisticated deepfake technologies. The insights presented aim to bridge existing gaps in research and pave the way for innovative solutions in combating the misuse of deepfake technology.

The rest of the paper is organized as follows: Section II presents a recent literature review focusing exclusively on articles published in 2024. Section III covers a detailed explanation of the methodology utilized in collecting this systematic literature review. Section IV exhibits a background on deepfake technology followed by Section V which covers a variety of deepfake video detection approaches, classifying them into visual, audio, and multi-modal methods. Section VI provides a comparative analysis of the detection methods, emphasizing their efficacy, generalizability, and validity. In the end, Section VII summarizes the key findings of the paper and pinpoints the necessity for future advancements in deepfake video detection technology.

2. Literature review

The advancement of Deepfake technology coupled with the swift dissemination and broad accessibility provided by social media platforms, enables the rapid reach of these videos to a vast audience, generating significant public apprehension and unease [24]. Extensive alterations to visual attributes, including facial expressions, age, and other physical characteristics, are enabled by AI-powered applications such as DeepFaceLab [25], Face Swap [26]. The primary contributions of this literature review include a thorough examination of existing research on deepfake video detection technologies, with a specific focus on studies published in 2024 within this section. An in-depth analysis of the most recent deepfake video detection methods has been conducted, including detailed performance evaluations of these techniques. Additionally, the challenges encountered in the field of deepfake video detection are explored in Section IV below.

Stroebel et al. [27] discuss many contemporary deepfake video detection technologies, assess emerging trends, and pinpoint future research avenues. The documents reviewed predominantly concentrate on video, image, or both types of media, with only a small fraction addressing audio or mixed media. Numerous machine learning models utilizing a GAN framework are examined, though the focus is primarily on deep learning architectures based on artificial neural networks, such as convolutional deep learning networks, deep neural networks (DNN), and long short-term memory (LSTM) networks. Several models are highlighted, including MSTA_Net, which detects anomalies by analyzing the entire texture of images, along with a self-supervising decoupling network and an integrated texture/noise framework. Advancements in synthetic voice generation and conversion for audio deepfake video detection are made. Techniques for detecting audio deepfakes are compared, revealing that image-based classification methods utilizing deep learning algorithms outperform feature-based approaches in performance [27].

Cunha et al. [28] suggested the utilization of a Particle Swarm Optimization (PSO)-based method, which is derived from bird foraging, for boosting deepfake video detection. The participants adopted EfficientNet pre-trained on ImageNet, exposing them to deepfake datasets to discern particular patterns in the frames of the videos. Another version of EfficientNet was used with a Gated Recurrent Unit (GRU) for the classification of the video sequences. In setting the hyperparameters for both models, the PSO algorithm was applied, including leader signals and reinforcement learning as the exploration. The experiment was based on 1016 real and 8425 fake videos from the training dataset of the DFDC, as well as a test dataset of 206 real and 1636 fake videos. The proposed models proved to be better than previous techniques, in the identification of falsifications in videos [28].

Su et al. [29] proposed a novel method known as the Unified Forensic Scheme by Content Consistency (UFCC). While Deepfake technology is frequently employed to alter the appearance of individuals in videos, UFCC can identify Deepfake content by comparing modified segments in images with the corresponding video content. The methodology assumes that alterations create discernible differences between edited regions and intact areas within the visuals. This approach applies to both photographs and videos. The authors acknowledge the challenge of assembling a diverse supervised learning dataset due to varying tampering techniques. To mitigate this issue, they utilize information from original, unaltered content during training instead of relying solely on modified data. A neural network is trained for feature extraction and classification of image patches, while a Siamese network is employed to evaluate the similarity between patch pairs. For Deepfake video detection, the analysis primarily focuses on facial regions, estimating authenticity by comparing facial similarities across consecutive frames. The UFCC scheme outperforms existing methods and distinguishes itself by providing a comprehensive detection framework that effectively addresses both photographs and video manipulation scenarios [29].

Chen et al. propose a new approach for detecting deepfakes in [30] using 3D spatiotemporal trajectories. A reliable 3D model has been incorporated to track motion characteristics from both 2D and 3D frames which helps in controlling areas like large head movement or low lighting. Also, facial expressions are decoupled from head movement and a sequential analysis method is used for real and fake face distinction. The testing of the multiple numbers of compressed deepfake samples justifies that this method works efficiently and has profound practical application [30].

The advancements in deep learning and big data have led to the development of popular deepfake technologies that allow the generation and editing of faces in images as well as in videos. To solve these problems, this paper proposes a new closed-loop detection framework referred to as ReLAF-Net. It uses a restricted self-attention mechanism which can decide the usage of self-attention to deep CNN features, and improves the learning of the local relations and dependencies at both fine level and global level. This attention mechanism is very flexible in design and is very easy to plug into CNN networks and improve the detection rate. Moreover, an adjustable regional frequency attribute extraction method is proposed to analyze RGB images for detailed frequency domains that help to extract forgery indicators. When tested on the high-quality FaceForensics+ + dataset, ReLAF-Net achieved a detection accuracy of 97.92 %, surpassing other methods [31].

Altogether, this work proposes a fresh approach to boost the localized attention features by analyzing multi-scale spatial-frequency relations and greatly improving the method of detecting deepfake videos. The model that the authors propose further improves generalization and, by employing the use of restricted self-attention and a feature extraction algorithm with high-frequency detail, helps to identify tiny indications of forgery [31]. This approach effectively addresses the problem of finding out the advanced deepfake manipulations and it is also more effective than the existing methods in recognizing the altered images and videos. Nevertheless, one has to be aware of the limitations of the model. It stands tall in most testing scenarios and could be less effective with extremely high-quality deepfakes, or deepfakes created out of other state-of-the-art methods [31].

The emergence of deepfake has become rampant and has proven difficult to detect fake videos leading to problems such as the spreading of fake news, identity theft, and infringement of privacy. New approaches for detecting deepfake videos have been proposed by [32] using the integration of ant colony optimization–particle swarm optimization (ACO-PSO) with deep learning features. The presented approach of feature extraction obtains ACO-PSO features from the spatial and temporal domain of the video frames and models the subtle patterns that reveal the deepfake manipulation. This is done to establish a basis by training a deep learning classifier that must distinguish between a real video and a deepfake video. A very large number of experiments underlying the method show its advantage in terms of detection accuracy, robustness to various manipulations, and compatibility with new data. The method used in the research achieved an acceptable level of accuracy, which is equal to 98. 91 %. The integration of the ACO-PSO features with the deep-learning algorithm improves the analysis, making it more precise and reliable every time in detecting deepfake content which helps to address the issue of deepfake content, thus preserving the credibility of digital media. This approach stands to benefit from larger sample sizes but produces fairly good results even with a fewer number of picture samples [32].

Computer-generated content is increasingly being used to create highly realistic fake videos that can easily fool viewers. GANs are primarily responsible for this technology, as they have the capability to produce such convincing fake content that it often goes undetected by traditional detection methods. Various studies have shown that convolutional neural networks (CNNs) are commonly used to detect and identify altered videos. However, these approaches primarily focus on the spatial features of individual frames, often overlooking the crucial temporal information derived from the interactions between frames. To tackle this issue, Reddy et al. [33] have introduced an optical flow-based feature extraction method that captures the temporal dynamics necessary for effective classification. Furthermore, our model has taken this approach a step further by combining recurrent neural networks (RNNs) and CNNs with the optical flow architecture. This integration has allowed for a more comprehensive analysis of video content, enhancing our ability to detect and classify altered media accurately [33].

The systematic review that was conducted by Heidari et al. [34] has noted that conventional deep networks are widely used in detecting deepfakes in videos and considerable shortcomings in existing approaches were outlined by the authors. CNNs and recurrent neural network-based deep-learning frameworks have been critically examined for deepfake video detection with details on aspects of strength and weakness, on issues related to accuracy, robustness, and complexity. As noted, it has been a common concern that specific detection tools are assessed on accuracy and their performance, but not much use has been made of key functions such as security. Even though DL can solve these problems, some factors such as inadequate access to non-English materials and lack of detailed description of algorithms in some research have been encountered. Still, the findings of the study will help to establish more effective approaches for creating real-life DL-based deepfake video detection technologies as well as future multi-stakeholder cooperation [34].

DeepFake technology, a fusion of "Deep Learning" and "Fake," manipulates facial data in videos, creating convincing but false images and videos using advanced methods like variational autoencoders (VAE) and GANs [35]. Detection of DeepFakes is often approached as a binary classification problem. Traditional methods analyze facial features or physiological signals, while deep learning techniques use CNNs to extract and classify features. Although methods like the Xception model and various multiscale or pixel-region networks have shown promise, their accuracy can be compromised by video compression and the blending of manipulation with compression artifacts. To address these issues, this paper presents a Multi-Scale Interactive Dual-Stream Network (MSIDSnet). This new network uses a multi-scale fusion (MSF) module to capture facial features across different domains and an interactive dual-stream (IDS) module to integrate these features effectively. The network processes video frames for spatial domain information and high-frequency noise for frequency domain information, improving detection efficiency and accuracy. Vision Transformer (ViT) is used to fuse features from both domains, enhancing the network's ability to detect DeepFakes [36].

To summarize the findings of the literature review, we have made Table 2, which contains the key themes and findings of the papers that are reviewed for this study.

3. Methodology

The guidelines provided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) have been followed to conduct the procedure of systematic literature review highlighting the detection method for identifying deepfake videos. PRISMA is selected over other methods because of its clarity and transparency and its ability to reduce bias by promoting comprehensive reporting. The procedure comprised of the following key steps: identification, screening, eligibility, and inclusion of studies, as shown in the PRISMA diagram (see Fig. 1). Among the various data reviewed, the studies from 2023 and 2024 were predominantly included, reflecting the latest advancements in the field.

3.1. Identification of studies

Google Scholar has remained the primary source for extracting relevant articles. Initially, 103 articles were added to the list; no other results were recovered from other databases. The possibility of data

Table 1
Research Questions.

No#	Question	Purpose
RQ1	What are the main categories of deepfake video detection methods?	The main methods used in deepfake video detection include visual-based methods, audio-based methods, and approaches that utilize both visual and audio signals simultaneously.
RQ2	How effective are these detection methods in real-world scenarios?	Assess the robustness and accuracy of detection techniques across varying conditions.
RQ3	What challenges do current deepfake video detection methods face?	Analyze the limitations, such as adversarial attacks and generalization issues, in detection techniques.
RQ4	How can detection techniques be improved to address these challenges?	Propose improvements and future research directions for enhancing detection robustness and generalizability.
RQ5	Are there trends or advancements emerging in deepfake video detection?	Identify new methods or innovations in deepfake video detection to adapt to evolving technologies.

Table 2
Findings from Literature review.

Aspect	Key Findings	References
Deepfake Risks	Realistic videos created for malicious purposes like blackmailing, misinformation, and privacy invasion. Social media amplifies their reach and impact.	[24], [25], [26], [87], [88]
Challenges in Detection	Difficulty in distinguishing real and fake media due to advanced AI tools like DeepFaceLab, Face Swap, Zao, and Reface.	[24]
Detection Techniques	Various models like CNNs, DNNs, and LSTMs; frameworks like MSTA_Net and texture/noise-based approaches.	[27]
Audio-Visual Approaches	Audio deepfake detection methods outperform feature-based approaches.	[27]
Particle Swarm Optimization	PSO-enhanced EfficientNet with GRU improves deepfake detection by identifying patterns in video frames.	[28]
Unified Forensic Scheme (UFCC)	Detects deepfakes by comparing modified regions in images/videos to original content; addresses dataset variability.	[29]
3D Spatiotemporal Trajectories	Tracks motion characteristics in 2D/3D frames; addresses challenges like lighting and head movement.	[30]
ReLAF-Net Framework	Uses restricted self-attention and frequency domain analysis; achieves 97.92 % accuracy on the FaceForensics+ + dataset.	[31]
ACO-PSO Integration	Combines ant colony optimization and particle swarm optimization for spatial-temporal feature extraction; achieves 98.91 % accuracy.	[32]
Optical Flow Methods	Captures temporal dynamics with RNN-CNN integration for enhanced classification.	[33]
Systematic Review	Highlights strengths and weaknesses of CNN and RNN models; identifies gaps like inadequate datasets and lack of multi-lingual materials.	[34]
MSIDSnet Framework	Employs multi-scale fusion, dual-stream modules, and Vision Transformer (ViT) for robust deepfake detection.	[36]
Future Challenges	Generalization issues, real-world adaptability, compressed video challenges, and evolving deepfake techniques remain critical.	[27], [34], [36]

duplication was eliminated by clearing away 11 articles leaving behind only 92 exceptional articles for examining.

3.2. Screening process

All 92 records have been explored and screened including the title and theory per article. 11 articles were deemed irrelevant to the topic of deepfake video detection, leaving behind 81 research papers for examination.

3.3. Eligibility criteria

3 reports did not qualify as they were deemed irretrievable. Again 78 reports were tested for eligibility criteria and 5 were classified as peripheral and irrelevant to the research topic. Research papers that were more directed towards the use of AI techniques rather than highlighting the detection methods or possessed insufficient and vague empirical data were subject to exclusion.

3.4. Final inclusion

Eventually, after following a rigorous procedure of examining the research papers on the established criteria, 73 reports became part of the review. These studies encompassed details that gave a thorough understanding of a variety of techniques and issues with deepfake video detection methods. Once again, it was ensured that mostly recent studies from the years 2023 and 2024 are part of the review to make it easier to discern the latest developments in the field and pinpoint potential flaws.

4. Background on deepfake technology

4.1. Overview of deepfake technology

Deepfake technology in this domain pertains to the usage of artificial intelligence (AI), particularly deep learning techniques, in creating synthetic but realistic media, where a person in an already existing image or video is replaced with the likeness of another. This technology makes use of GANs to create extremely realistic content that can imitate facial and other expressions, voices, and gestures convincingly from real people's faces. The application of GANs in this field has accelerated the production of deepfakes, whereby manipulated content of high quality can be produced without requiring high technical skills [37].

These improvements in the creation of deepfakes have huge implications, some positive and others negative. While this technology has been put in place in creative industries such as the production of films and video gaming, misuse of the same technology has raised a lot of concerns. Deepfakes have been weaponized not only in distributing political disinformation and in identity theft but also in creating non-consensual explicit content—a fact that has raised widespread alarm about its potential impact on privacy and security. That problem is further compounded by the ease of availability of deepfake generation tools, placing the production of convincing forgeries within the capability even of individuals with basic technical knowledge [38].

4.2. State-of-the-art deepfake algorithms

While deepfake video generation algorithms have recently seen a lot of development with advances in deep learning, computer vision and probabilistic modeling, altogether. In the Table 4, the state-of-the-art practices for video deepfake generation are summarized. These are since frameworks like Generative Adversarial Networks (GANs) where two neural networks, the generator and the discriminator, compete against each other to generate realistic synthetic data, such as images or text. The generator creates fake data, and the discriminator tries to distinguish between real and fake data. Through this competition, GANs can produce high-quality, realistic outputs. In addition, Variational Autoencoders (VAEs) take advantage of to create really realistic visual material. Moreover, other tools such as DeepFaceLab and Face2Face provide a level of accuracy in face manipulation and edit expression. However, to produce more authentic and flexible 3D outputs, these methods introduce more control over geometric and expressive detail with NeuralTextures and 3D Morphable Models. Furthermore, these algorithms are made even stronger with more advanced datasets like CelebA and FaceForensics+ +, tackling severe problems such as variations of lighting, occlusions and motion artifacts.

4.3. Challenges in deepfake video detection

Despite the considerable progress in deepfake video detection, there are a lot of challenges. The most important one is that the techniques used to generate deepfakes change at an extremely fast pace; thus, they keep introducing forgeries that are getting closer to real videos and becoming more difficult to detect [39]. Also, the quality of deepfake videos would most definitely vary according to the resolution, compression level, and even the platform on which they are being shared. Therefore, a single detection methodology may not be able to retain precision for all sorts of scenarios. Moreover, attacks against the detection systems have revealed some adversarial ones. Specifically speaking, this sets a demand for resilience in the approach able to

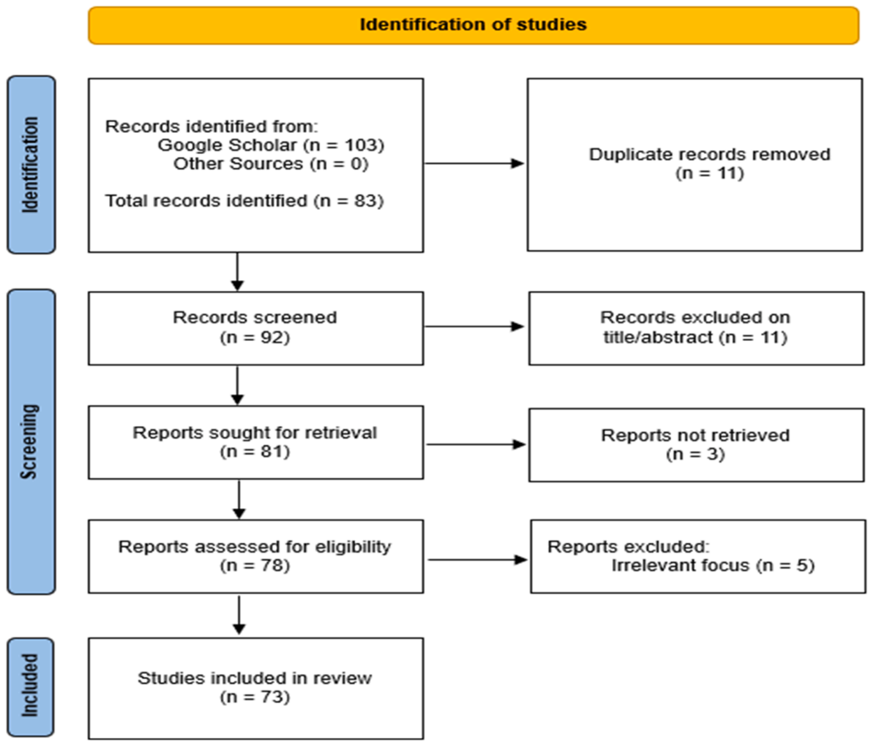

Fig. 1. Prisma diagram.

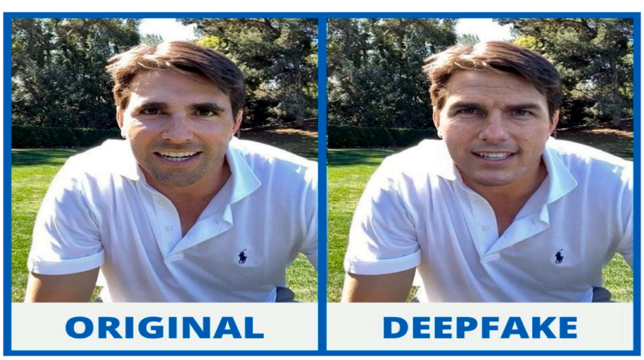

Fig. 2. Original vs deepfake. [89].

withstand manipulation intended to avoid detection [40]. Research shows that ninety-six percent of organizations have challenges in data quality as well as labeling within deep learning projects. Therefore, deepfake video detection, which depends heavily on deep learning approaches, has the same problems [41].

The quality of training data is crucial for model performance. Many deep learning-based detection methods are computationally demanding and require significant resources. Efforts are underway to develop data-efficient models that reduce the need for extensive training data. The effectiveness of DL techniques often necessitates large datasets, which can be costly and time-consuming to obtain. Deepfake video detection methods are also plagued by high computational costs, which complicate real-time applications and legal use [42]. The high resolution and temporal complexity of videos require substantial computational resources, potentially hindering timely detection and mitigation efforts. The reliability of current deepfake video detection methods is insufficient, especially when faced with new and evolving deepfake techniques. The lack of standardized benchmarks impedes the evaluation and comparison of detection methods. There is a scarcity of comprehensive benchmark datasets, which limits the ability to objectively

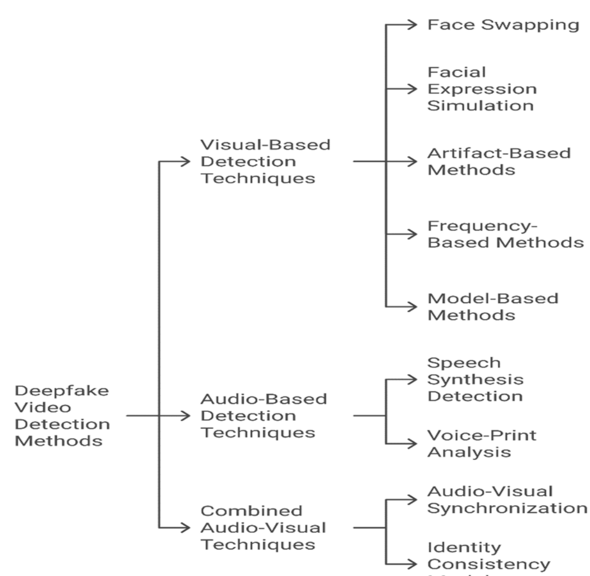

Fig. 3. Chart of Deepfakes video detection methods.

assess and improve detection methods. Efforts are ongoing to develop more robust benchmark datasets, such as the Deepfake video detection Challenge (DFDC) and Celeb-DF, to provide better evaluation standards. DNNs often lack robustness and perform poorly outside their training environments, making them vulnerable to adversarial attacks. Ensuring robustness is critical for deepfake video detection systems to maintain accuracy in real-world scenarios and withstand various adversarial threats [42].

5. Deepfake video detection techniques

Deepfake video detection methods have been categorized into three

Fig. 4. Deepfake artifacts [89].

Table 3
SLR Search Strategy.

Type	Description
Aim	Gather existing literature that outlines deepfake video detection methods, focusing on visual, audio, and multi-modal approaches across diverse datasets and techniques.
Data Sources	Articles were primarily located in databases such as "Google Scholar", "IEEE Xplore", and "SpringerLink", published between 2023 and 2024.
Query	A keyword search using the following query: "deepfake video detection" OR "video forgery" OR "face swap" OR "fake audio" OR "GAN-based detection" AND "real-world datasets". Articles were limited to those from 2023 and 2024 to ensure the inclusion of the latest developments.
Method	Using the search query, 103 articles were initially identified. Duplicates were removed, leaving 92 for screening. Titles and abstracts were reviewed to assess relevance. After exclusions, 73 articles were selected for full review and analysis.
Inclusions/ Exclusions	Studies published in 2023 and 2024 focus on deepfake video detection techniques. Books, articles with insufficient technical details, and older publications (before 2023) were excluded.

main areas: visual-based, audio-based, and multi-modal approaches or deepfake video detection using visual and audio signals simultaneously. Each type addresses different facets of deepfake manipulation, and the goal is to detect inconsistencies that arise in either visual, audio, or the combination of both modalities in fake media.

5.1. *Visual-based detection techniques*

Visual-based detection techniques primarily focus on identifying artifacts or inconsistencies in the visual features of deepfake videos. Various deepfake methods are currently employed, including identity replacement, where one person's face is substituted for another's, and expression alteration, which modifies facial expressions in videos. Different methods for identity replacement are used by tools such as FaceSwap, Faceswap-GAN [43], and DeepFaceLab. Recent progress in deep learning has led to significant improvements in the detection of these alterations, especially with ConvNets. For instance, anomalies in modified videos are detected by the mesoscale convolutional neural network, while geometric and channel-wise relationships are captured proficiently by the extreme inception network, though it encounters difficulties with fine details. A novel method proposed by Essa et al. [44] has been developed that fuses features from three transformer-driven visual models: Dual-attention transformer frameworks, inception architecture transformer, and vision transformer with group propagation. Local and global contexts are captured by the Dual-Attention Transformer, varied visual frequencies are handled by the Inception

Table 4
State-of-the-art Methods.

Technique	Description	Key Features	References
Generative Adversarial Networks (GANs)	Framework with generator and discriminator working iteratively to create realistic content.	CycleGAN for domain adaptation, StyleGAN for high-resolution facial synthesis.	[20], [33]
Variational Autoencoders (VAEs)	Probabilistic models that learn latent space representations for image and video synthesis.	Encoder-decoder architecture for data reconstruction and manipulation.	[21], [25]
Face2Face	Real-time facial reenactment allowing control over target expressions and speech.	Uses a blend of computer vision and deep learning to modify expressions in real time.	[27], [28]
DeepFaceLab	Deep learning tool for face swapping in videos.	Implements encoder-decoder models for seamless face exchange in video frames.	[24], [29]
NeuralTextures	Synthesizes high-resolution textures for realistic facial manipulation.	Precise control of geometry and expressions in generated videos.	[30], [32]
3D Morphable Models with DL	Combines 3D modeling with deep learning for detailed facial synthesis.	Enables fine control over facial geometry and expression with 3D enhancements.	[31], [33]
Advanced Datasets	Rich datasets like CelebA, FFHQ, and FaceForensics+ + used for model training.	Address challenges like lighting, occlusions, and motion blur, aiding model development.	[22], [34]

Architecture Transformer, and high-resolution details are maintained by the Vision Transformer with Group Propagation. These features are integrated by a multi-layer perceptron mixer to enhance deepfake video detection. The method has been tested on the FaceForensics+ + and Celeb-DF datasets, where an AUC score of 99.84 % was achieved on Celeb-DF, demonstrating its robustness and broad applicability compared to other leading-edge technologies. Visual-based detection techniques can target both low-level artifacts such as pixel irregularities or high-level anomalies in facial expressions [44]. Visual-based detection techniques are divided into five sub-categories which include artifact-based methods, face swapping, facial expression simulation, frequency-based methods, and model-based methods. Each type employs distinct techniques for detecting anomalies:

Face Swapping: A procedure involving the exchange of faces between individuals in videos was highlighted by Yu et al. Early work in this field was initiated approximately six to seven years ago, during which high-quality face-swapped images were generated from static photos using convolutional neural architectures. Although temporal continuity for video was not achieved by these early methods, subsequent advancements have led to improvements in their effectiveness [45]. For instance, in 2017, a novel method was introduced for creating videos using only one red-green-blue (RGB) image and a source video. Textures were adjusted frame by frame by employing a deep learning network, with features from the target image applied to the source video. As a result, the target face could be superimposed onto the source video, replacing the original face. In the same year, the first widely known face-swapping video created using this method was posted on social media platforms, causing considerable surprise [46]. It is widely accepted that the development of deepfake algorithms was inspired by earlier work in which convolutional neural architectures were employed for face swapping. This inspiration led to a global surge in face-swapping videos, produced for both positive and negative purposes. An improved version of the original deepfake algorithms, known as the Face

Table 5
Summary of recent papers and data sets from 2024.

Ref.	Year	Object	Target	Benefits	Limitations
Heidari et al. [34]	2024	Delivering an extensive overview of research on deepfake video detection methods utilizing deep learning algorithms.	DL-ML techniques	The benefits include a systematic analysis of various detection algorithms and tools and a thorough examination of their performance metrics.	limitations include challenges such as inaccessible non-English resources, inadequate algorithm descriptions, and the exclusion of relevant studies, which hinder comprehensive evaluation and comparison of detection methods.
Cunha et al. [28]	2024	The research aims to enhance deepfake video detection by using advanced deep learning models and a new Particle Swarm Optimization (PSO) algorithm to improve accuracy in identifying manipulated videos.	Hybrid EfficientNet-GRU and EfficientNet-B0-based models.	The proposed models show improved performance in detecting deepfakes compared to previous methods, thanks to the optimized PSO algorithm and advanced deep learning techniques.	The study may face challenges related to the adaptability of the models to new types of deepfakes and the potential limitations in generalizing across diverse datasets and video qualities.
Reddy et al. [33]	2024	The object of the study is to enhance the detection of "deepfake" videos by integrating optical flow-based feature extraction with RNNs and CNNs.	**RNN and CNN**	The study improves deepfake video detection by integrating optical flow-based feature extraction with RNNs and CNNs, allowing it to capture and classify time-related features more effectively than traditional methods.	The approach may require substantial computational resources and might still struggle with deeply complex or novel deepfake techniques not yet encountered.
Gao et al. [31]	2024	The objective of the study is to improve deepfake video detection by analyzing both audio and video together over time	bimodal temporal feature prediction.	The benefits include improved accuracy with 84.33 % success and enhanced detection through temporal analysis.	Limitations include basic audio analysis and a lack of real-time detection capability.
Su et al. [29]	2024	IFCI aims to detect tampered areas in still images and identify Deepfake videos by analyzing content consistency.	CNN for feature extraction and a Siamese network	IFCI provides a unified approach applicable to both still images and videos, avoids the need for extensive tampered datasets, and demonstrates superior accuracy and stability in detecting manipulations.	It struggles with severely overexposed areas and compressed or low-quality images, and the model's performance might be affected by the lack of training data with controlled image degradation or compression.
Cheng et al. [36]	2024	The study aims to enhance Deepfake video detection by developing the MSIDSnet, which integrates spatial and frequency-domain features using a ViT.	MSIDSnet and ViT	The model achieves high accuracy on both high-quality (99.30 % on Celeb-DF-v2) and low-quality datasets (95.51 % on FaceForensics++).	Limitations involve potential challenges with generalization to unseen datasets.

Substitution Generative Network, was introduced to enhance the realism of faces [46]. Improvements in GAN-facilitated face creation and facial feature modification have been made. Despite these advancements, identity mismatches continue to be a challenge, with many methods relying on complex models and numerous loss functions to address issues related to facial geometry and identity characteristics [47].

In comparison to these strategies, DeepFace Technology (DFT) is proposed by Liu et al. [48] as a multi-angle face-swapping solution, a system that is designed to swap faces using multiple images. A large amount of data from both the person being swapped and the person being replaced is utilized by DFT. Hundreds of portraits of the person being swapped, displaying various angles and backgrounds, are collected. This data is used to provide detailed identity information. Neural architecture is then instructed to acquire and utilize this identity information. With this approach, highly realistic face-swapping results that resemble cinematic quality are produced by DFT. Ethical concerns are raised by face-swapping technology, which enables the creation of fake images and videos [48]. The widespread use of these face-swapping techniques can impact public discourse and infringe on personal rights, making the detection of such manipulations a crucial challenge. Various detection methods, including Pixel-Resolution Recurrent Networks and Dual-Stage Deep Neural Networks, have been proposed, along with techniques focused on identifying tampering traces. In addition to detection, improvements in generation methods are being addressed. To support detection efforts and raise awareness, the publicly available DFT project has been established. This project aids in the creation of high-quality face-swapping videos and contributes valuable data for forgery detection research [48].

Facial expression simulation: Facial expression simulation is differentiated from face swapping by its focus on controlling facial expressions in videos, allowing for the creation of videos where individuals perform actions or expressions that were not originally made. The first algorithm for face reenactment is traced back to the early 21st century. A facial prototype, which can be altered to reflect various expressions, is proposed by Thies et al. [49]. Many subsequent methods are built upon this concept, with models being used to adjust facial images. Although very realistic faces are generated by these techniques, issues with temporal consistency are often encountered. Recently, significant advancements in face reenactment research have been achieved due to the increase in computing power [49].

Artifact-based methods: Artifact-based methods aim to identify the subtle visual artifacts introduced by deepfake generation algorithms. These artifacts might include issues such as inconsistent lighting, unnatural facial textures, or mismatched shadows, all of which result from the limitations of current deepfake generation techniques. Additionally, discrepancies in eye movement and unnatural blinking patterns have been effective markers in identifying deepfakes [50]. Deepfakes are significant due to their ability to generate plausible yet fake content that is often indistinguishable from real media. Since their emergence, considerable effort has been devoted to combating and detecting media forgery, resulting in the development of various approaches. In the absence of a universally effective Deepfake detection method, Bansal et al. [50] suggest that artifact-based techniques show great promise. This study aims to evaluate existing methods for detecting deepfake exploits in order to identify the most effective approach for this task, based on parameters such as convergence/ approach, precision, and equal error rate (EER).

Frequency-based methods: Despite the notable advances in deepfake video detection, many challenges remain. Probably the most relevant one is that the deepfake generation techniques are changing at a very fast pace; thus, they keep introducing forgeries that get closer and closer to real videos; hence, it will become more difficult to detect them. Furthermore, the quality of deepfake videos would certainly differ depending on resolution, compression level, and even dependent on the platform on which they would have been shared. Thus, one methodology of detection perhaps would not retain precision for all sorts of scenarios. Besides, attacks against the detection systems revealed some adversarial ones. More specifically, this places a burden on the approach to be robust against manipulating signals to avoid detection [51]. A new detection method was introduced by Malik et al. [51], based on the DFDC and Face Forensic datasets. During the preprocessing stage, frequency-based frame extraction was employed for each video. The approach integrates CNN and LSTM techniques, resulting in an accuracy of 82 %. The primary goal of this work is to assist researchers in detecting fake videos using DL techniques [51].

Fig. 5 displays the frequency domain analysis of extremely compressed facial images: a) a real facial image and b) an imitated face image. The lookalike image (b) lacks the texture and detail of facial features that are evident in the real image (a), particularly in the areas of the nose and mouth.

5.1.1. Model-based methods

As deepfake generation methods become more advanced, the need for systems that can tell the difference between real and fake images is increasing. This challenge isn't limited to images; it also affects text, with recent studies analyzing deepfakes in tweets to spot fake content on social media. To address the problem of detecting deepfakes in videos, many video-based detectors have been created. While some methods analyze the changes over time in manipulated videos, most approaches focus on examining each frame individually.

Model-based methods rely on deep learning techniques, such as CNN, which automatically learns features from videos about deepfakes. These models are trained on huge numbers of both good and bad quality real and fake media with the ability to learn the artificiality introduced in deepfakes only. For instance, CNN can easily locate minute visual differences that the human naked eye or brain could hardly catch [52]. Further, hybrid models have also been proposed that amalgamate CNNs with LSTM networks to capture spatial and temporal discrepancies, respectively; performance is enhanced in the detection of deepfake videos that span several frames [53]. Dhar et al. [54] describe a new deep-learning approach that enables one to differentiate fake videos created by AI from real videos. The methodology has special effectiveness in detecting fake videos, particularly videos with changes in content. A ResNext-CNN is recognized as the main component of the system that detects important features of the frames of the video streams. Further, the extracted features are employed for training an LSTM Recurrent Neural Network (RNN). This kind of RNN is used in the identification of the content of the videos and to separate the deepfake ones from the real videos. To overcome this problem during modeling, it was tested on a large and balanced dataset so that the suggested model would have a realistic approach to real-time situations. This dataset amalgamated three other datasets; FaceForensic++, Deepfake Detection Challenge, and Celeb-DF datasets. It has been demonstrated that the system performs competitively, achieving effective results through a straightforward and efficient approach.

The ForgeryNet dataset, which is the largest and most diverse, has become a popular choice for deepfake detection research. EfficientNet, a type of CNN, is highly effective for this task and is used in many leading solutions, including the top entry in a deepfake detection competition. In recent years, new detection methods appeared with the emergence of Vision Transformers in the computer vision area. For example, there is an approach aimed at using Transformers combined with convolutional nets for analyzing faces' traits or at using an EfficientNet B7 model that has been trained and a Vision Transformer. Many kinds of innovative work have been done to merge different types of Vision Transformers, such as the Cross Vision Transformer and EfficientNet B0. EfficientNet has been further improved with the introduction of EfficientNetV2, which is designed for smaller models, faster training, and better performance on the ImageNet dataset [55–59].

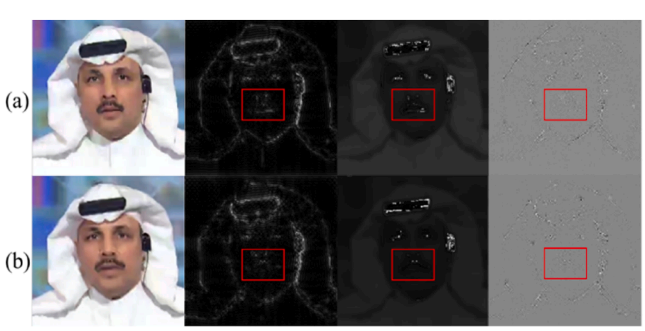

Fig. 5. Deepfake artifacts.

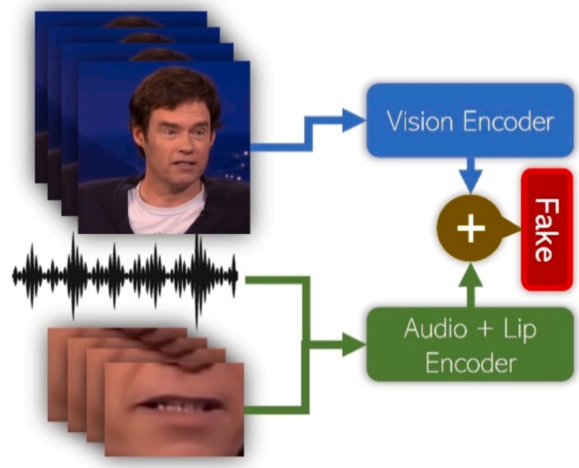

Fig. 6. Audio-visual synchronization.

5.2. Audio-based detection techniques

Audio-based deepfake video detection focuses on analyzing the audio tracks of videos, especially where voice manipulation is involved. This includes detecting synthesized speech or audio that is inconsistent with the corresponding visual content.

Speech Synthesis Detection: The generated speech in deepfakes usually contains subtle artifacts that can, with due care and appropriate analysis of the audio signal, be identified. Techniques such as spectrogram analysis, which represents audio frequencies as a function of time, can reveal unnatural speech patterns, which can relate to inconsistencies in tone, pitch, or rhythm. These inconsistencies sometimes originate because the tone-to-speech models are not perfectly trained—those that are generally used to create the audio in deepfakes. Such manipulations were more effectively detected by finding phoneme mismatches, which concern cases where lip movements are not fitted to the speech produced [60].

Voice-Print Analysis: The other strong technique in audio deepfake video detection involves something called voice-print analysis. This method focuses on a person's voice features, which include frequency, tone, and inflection. In running the voice-print analysis, these features would be compared with known samples of the speaker to find manipulation or synthesizing of the audio [61].

Deepfake Video Detection Using Visual and Audio Signals Simultaneously/Multi-modal detection techniques (Combined Audio-Visual Technique)

Multi-modal detection techniques enhance the robustness of deepfake video detection by integrating both visual and audio signals. This combined analysis allows for the identification of inconsistencies that might be missed when focusing solely on one modality. By examining the interplay between audio and visual cues, these methods are better equipped to detect subtle manipulations, as distortions in one modality can be exposed by cross-referencing the other. For the purpose of detecting intricacies in contrived deepfake scenarios, this hybrid approach boosts the efficacy of detection systems [62].

Rehman et al. [63] discussed the problem of identifying suspicious behaviors in outdoor zones, which are traditionally observed by surveillance cameras to watch over the public. It presents a novel approach for automatically detecting such abnormalities where both sound and video inputs are used. The detection algorithm uses visual features like optical flow and techniques from particle swarm optimization (PSO); sound characteristics including energy and MFCCs are also employed. These combined features give rise to the generation of an inference system. The performance of this method is evaluated using public datasets associated with UMN and their audio files. In this case, the use of this algorithm reveals a high level of accuracy when compared to other advanced algorithms [63].

Audio-Visual Synchronization: The assessment of the synchronization between a video's visual and audio components is one of the potent strategies of the model. A contrived video might not exhibit a huge level of correlation between the lip movements and the speech. This is one of the areas where deepfake technology lags behind as it often struggles to create harmony between the audio and video content. Subtle temporal disarrays can be checked for synchronization between the two modalities by the models, created by the researchers, trustable deepfake detecting indicators [64].

Identity Consistency Models: The contrived deepfake visuals are crafted with the extraction of biometric facial attributes from videos and audio. The Multi-modal approach identifies the discrepancies in the sample videos provided of the same individual. Any disparity—such as a mismatch between a person's voice and their facial expressions—indicates the clue against the existence of deepfake [65]. Deepfake is a composition of high-quality audio and visuals but often the model can identify the variability and hint of alterations in one or another [65].

Adversarial Attacks and Defense: The advancement in deepfake video detection techniques doesn't hinder the growth in deepfake generating tools. More advancements have been made in deepfake creation techniques to keep the alterations as subtle as possible, tricking the detection models into misidentifying the deepfakes as the original video. The detection models are inculcated with adversarial training on real-life scenarios to strengthen their ability to distinguish these subtle discrepancies in the deepfake videos and reduce the chances of inaccuracy in the discernment. Hence, Detection systems with these defense mechanisms are well-built for combating the subtle measures taken to evade conventional detection methods [66].

6. Comparative analysis of detection methods

Cutting-edge advancements in data-driven learning and automated intelligence are being leveraged across various fields, such as criminal investigation, the medical field, digital helpers, information security, and automation technology. However, the potential for misuse exists, as these technologies can be exploited to create convincing fake news that spreads misinformation and erodes public trust [67]. Audio or visual content that has been artificially created or altered using advanced neural architectures is referred to as deepfakes. While positive applications of deepfakes, such as the enhancement of publication front-page pictures, are recognized, they can also be employed for harmful purposes, such as superimposing faces onto inappropriate content or fabricating audio recordings of public figures to discredit their standing [68]. The creation of sophisticated deepfakes has been facilitated by the widespread availability of mobile devices and user-friendly editing tools, posing challenges for current forensic technologies and investigators. Existing databases of picture, film, and sound deepfakes are reviewed in this paper, with the aim of enhancing the precision, flexibility, durability, and clarity of deepfake video detection techniques. Current obstacles and possible avenues for investigation in both deepfake creation and prevention are also examined. The support of scientists, developers, and professionals in comprehending and advancing sophisticated deepfake technologies is the intended goal [69]. In recent years, various methods have been developed in the realm of face deepfakes, including techniques for identity exchange, performance recreation, feature modification, and complete face generation, as explained by Akhtar et al. [70]. Similarly, methods for voice replication, speech generation, voice transformation, sound editing, language conversion, and segmented deepfakes have been established in the audio domain. Despite notable advancements, adaptability, immediate identification, and defense against malicious attacks are being struggled with by many detection systems for audio and video deepfakes due to their responsive nature. This ongoing struggle is being represented as a continuous arms race between those who are creating deepfakes and those who are trying to detect them. The latest developments in deepfake technology must be kept updated by researchers and practitioners to stay ahead in this critical contest [70].

The following is a detailed description of the attributes of detection models in terms of their effectiveness, generalizability, and robustness. The descriptions embody both potential influence and flaws in each method across numerous data sets and alteration techniques.

6.1. Effectiveness

In this section, we explore various deep learning approaches to determine which one is most effective technique for the detection of deepfake videos. This is important because video modification does not necessarily result in the same attributes as image modification, which makes it rather interesting to find out how different models treat videos. For instance, the research [71] focuses on three architectures: EfficientNet V2 which is a convolutional network, a normal Vision Transformer, and a Swin Transformer which is a fusion of CNN and transformer. In this study, two settings were considered by Coccomini et al. [71] to understand how well the proposed and investigated deep learning models generalize in identifying deepfake videos. In the first configuration, a limited training sample consisting of precise manipulation procedures was incorporated; in this case, EfficientNetV2 was found to outperform the other models. The Swin Transformer also demonstrated good performance, as indicated by the generated results. Based on the study's findings, it was concluded that the Vision Transformer is most suited for real-world deepfake detection when large datasets are available. However, it was suggested that EfficientNetV2 could be utilized when training data is limited [71].

Results show that the Vision Transformer excels at generalizing in cross-forgery situations, while the Swin Transformer performs well across different datasets. The attention mechanism helps these models in learning about deepfakes when the models are trained on mixed data. For a few data samples, the Vision Transformer has a hard time as opposed to EfficientNet V2 and Swin Transformer, which continue to excel when tested in such cases. The most recent development that makes it easier to forge images and videos through the process referred to as deep learning is termed deepfakes. Although there are some existing detection systems, mature learners complain that such systems fail to provide real-life examples, especially when new techniques are used. In this work, different deep learning models are explored to find which can better identify deepfakes. The findings suggest that CNNs excel with small datasets and specific manipulation techniques. On the other hand, Vision Transformers exhibit better performance in varied data sets and better generalization. Swin Transformer is data efficient and robust across datasets and is efficient with little data and fair on multiple datasets. Overall, attention-based models seem to offer better performance in recognizing deepfakes, especially in real-world situations [71].

The accuracy of deepfake video detection methods is reliant on the quality of the dataset used in conjunction with the specific alteration methods. CNN-based models are reliable in terms of generating accurate results and authenticity, particularly in controlled settings where efficiency in particularly identifying the modifications in facial features and light conditions. As mentioned earlier, the accuracy of these models relies heavily on the quality of datasets, therefore there are elevated chances of misidentification with these detection models. This accentuates the importance of variable and premium quality datasets for accurate detection. For example, CNN-based models have consistently reported high accuracies in controlled environments where they can detect minute inconsistencies in facial features, lighting mismatches, and skin textures in high-resolution images [72]. They are amazingly good at capturing pixel-level artifacts which may often be imperceptible to human eyes.

Further, the models that involve temporal analysis, which implement LSTM networks, perform well on inconsistency detection between successive frames of videos, which, again, is very crucial in identifying video sequence manipulations. All these developments have also brought forward the use of deep learning, though the drawback is in the fact that the deep learning models are not robust within real-world deepfakes in most low-resolution or compressed videos, expectable in social media platforms [73]. High-performance models that are usually trained on high-quality datasets suffer from various losses in accuracy when tested with low-quality content. Although there are small improvements using the frequency-aware detection methods and feature-based approaches further down the line, these methods also face a lot of challenges in bridging the performance gap between their performance in the quality of datasets.

Deepfake technology is advancing rapidly, creating significant challenges for detecting fake videos, which is crucial for maintaining the credibility of visual information. Conventional techniques are invariably plagued with problems such as how to select appropriate reference frames as well as how to extract correct features. A new model introduced by Wu et al. [74] is also dedicated to enhancing the detection by the incorporation of spatial and temporal elements, which helps to capture both the static details and the dynamic changes in video content. By doing the facial edge differences keyframe sequences are first defined so that the interesting parts of the video are concentrated. This step is important as the coordinates make it easier to pinpoint points in which these manipulations may be most prevalent. The sequences are then processed in two separate branches: one branch is designed to detect hidden artifacts that may indicate tampering, while the other branch examines inconsistencies that arise over time, such as unnatural movements. A self-attention mechanism is utilized to merge these features effectively. This approach allows for the dynamic weighting of the importance of different features, thereby enhancing the model's ability to detect subtle signs of forgery. Last, the fused features are introduced into the classifier, which makes the last decision on the fact that this particular video is real or fake. The results of testing conducted on established datasets like Faceforensics+ + and Celeb-DF confirm that this model not only improves accuracy but also demonstrates better generalization for different types of Deepfake videos in comparison with the existing approaches [74].

Many recent methods for detecting deepfakes analyze individual video frames, concentrating on visual details to assess whether the content is real or fake. While some techniques look at temporal inconsistencies in videos, the primary emphasis remains on spatial features. In this study [75], the authors present a new model that integrates spatial, spectral, and temporal information to improve the accuracy of fake and real video detection. It highlights how the Discrete Cosine Transform can enhance detection by capturing important spectral features from individual frames. This multimodal network achieved a detection accuracy of 61.95 % on the DFDC dataset. The study also tested several sub-networks to determine their effectiveness in deepfake detection. The LipSpeech model, which detected mouth motions and evaluated audio spectrograms, obtained an accuracy of 59.21 percent but was not very helpful, indicating that audio-visual synchronizations differ. In this research, FourierNet proved to be slow with an average accuracy rate of 50.20 % which explained a weak correlation between audio and visual spectral features. XceptionNet, combined with LSTM (Xception + LSTM), reached 54.82 % accuracy, but its focus on high-level features limited its effectiveness. However, the VSNet that integrated the feature of XceptionNet into the 1-D Discrete Cosine Transform (DCT) data of facial landmarks successfully raised the detection accuracy to 61.95 %. This highlights the value of incorporating spectral features in deepfake detection. This work proposed NOLANet, a multimodal network, intended to detect deepfake videos. The researchers did a good job trying to investigate the dataset of the Deepfake Detection Challenge and extracting numerous features. They introduced novel approaches for feature extraction in the frequency domain and introduced a model for audio-visual synchronization discrepancy in deepfakes. It was established that the most related sub-networks included LipSpeech which detected disparities in the temporal relationship of audio and video and VSNet using features from 1-D Discrete Cosine Transform (DCT) combined with XceptionNet results. This goes in support of the notion that spectral features can be included to improve deepfake detection [75].

6.2. B. Generalizability

Generalizability is pivotal for deepfake video detection methods that continue to be effective with the continuous evolution of new deepfake generation techniques. Techniques involving frequency domain analysis have exhibited superior generalizability. Because these approaches focus on the frequency patterns that are harder to manipulate, therefore allowing detection across a wide range of quality and compression levels [76], they are in a better position. Hybrid methods combining CNNs with temporal analysis models gain generalization by training on the spatial and temporal aspects of data. Models like these are capable of detecting inconsistencies across time, for example, mismatched facial expressions and speech. This makes them effective in detecting deepfakes across multiple frames [76].

Multi-modal methods, while speculated, stand against the typical one-dimensionality of approaches that rely on either of the factors by cross-referencing inconsistencies from the audio and visual components of deepfakes. Unfortunately, it is with these advances that adversarial attacks have proven to be a big test for generalizability. A lot of detection models have been found, which easily get fooled on adversarially manipulated inputs with the explicit purpose of exposing the weaknesses of the detection systems. The adversarial training approaches against such attacks are being devised and integrated into the deepfake detectors to improve generalizability [77].

7. Robustness

Deepfake video detection methods will only be effective in real-world applications if they maintain performance on diverse datasets and challenging conditions like compressed videos, occluded faces, and deepfakes with multiple subjects. Visual-based techniques, which are sensitive to pixel-level analysis, tend to perform well on high-resolution videos, detecting subtle deviations in facial features and lighting conditions. However, most of these methods struggle when applied to low-quality content, where compression artifacts and occlusion mask the visual anomalies that the models rely on. As a result, detection performance is significantly degraded in scenarios involving compressed or occluded media [78].

In contrast, frequency-based detection methods have shown greater robustness under similar conditions. These methods analyze the frequency domain to reveal subtle artifacts introduced during the deepfake generation process, even in heavily compressed or low-resolution videos. This makes them more robust compared to pixel-based approaches. Research has shown that frequency-domain analysis detects such subtle inconsistencies with high accuracy when applied to real-world deepfakes, where quality is often compromised for online distribution. Studies confirm that frequency-based methods outperform pixel-level models in challenging environments, providing a more reliable solution for detecting manipulated content in practical scenarios [79].

Additionally, multi-modal approaches have enhanced the robustness of detection by jointly analyzing both audio and visual data. These models evaluate the synchronization of lip movements with speech, adding another layer of detection. Even when a single modality, like the visual component, is convincingly manipulated, audio inconsistencies can still reveal the forgery [80]. Cross-referencing information between audio and video in multi-modal approaches improves the detection of deepfakes, ensuring that manipulations in one area can be identified through discrepancies in the other. This dual-layered approach is effective even in cases where one modality seems undistorted [80].

Adversarial attacks, however, pose a significant challenge to the robustness of detection systems. These attacks exploit vulnerabilities in detection models through subtle modifications introduced to bypass traditional detection mechanisms. Nevertheless, adversarial defenses have shown promising results in improving robustness. Adversarial training, where models are exposed to manipulated inputs during training, prepares detection systems to recognize and resist adversarially modified deepfakes [81]. This technique has been proven to enhance the resilience of detection systems, ensuring they maintain accuracy even against sophisticated deepfake generation techniques designed to evade detection.

8. Dataset variability and benchmarking

Deepfake video detection models significantly are trained and evaluated for the datasets that they have been exposed to. Most of the popularly available datasets, such as FaceForensics+ +, CelebDF, and DFDC, are dominated by high-resolution deepfake videos that have been created in highly controlled environments. Although having these datasets is important for many benchmarks, too often they fail to represent the full complexity and variability of real-world deepfakes. The deepfakes that are encountered on everyday usage, such as on social media platforms, are then of low quality and with more complex manipulation, for example, videos involving multiple subjects, occlusion, or heavy compression. There is a contrast between the performance of these models on controlled, premium-quality datasets and in real life. In controlled their efficiency surpasses their efficiency in real-world scenarios, hence making them less reliable for real-world use [82].

A dataset called DF-Platter is programmed to prompt real-world scenarios and provide an accurate picture of the difficulties deepfake video detection models may encounter. This dataset constitutes both high and low-resolution deepfakes, enclosing the scenes with altered subjects and accurately showcasing the variety in real life. DF-Platter possesses the technology to tackle both high and low-resolution videos, and indicate abnormalies like imploded faces. This makes it an important instrument for inquiring about the restricted abilities of the existing deepfake video detection models specifically their limited capacity when it comes to assessing lower-quality videos and detection of multiple alterations made in facial features [83]. The inclusion of this variability is crucial for testing the robustness and generalizability of detection models across diverse conditions. It's pivotal that these models possess the ability to assess variable datasets, high or low quality, only then they can be tested for robustness and generalizability.

Research highlights the importance of creating detection models that are strong beyond the walls of controlled experiments, as real-world scenarios are much more complicated, therefore these models must possess the ability to assess multiple different scenarios. Research utilizing DF-Platter and similarly, varied datasets has demonstrated the critical importance of evaluating detection models across a wide spectrum of scenarios, ensuring their effectiveness beyond the confines of controlled experiments [84]. If the detection models are extremely dependent on the high quality of datasets, then this might jeopardize the possibility of creating strong models that are clever enough to process the intricacies in real-world deepfakes. The solution is to incorporate a wide range of modifications in the datasets so they provide a more realistic and complex picture of real-world deepfakes enhancing the strengths of these models [85].

Detection models tested on one-dimensional scenarios shouldn't be deemed authentic and reliable for real-world functioning as the real-world scenarios might not be as simple. Instead, multidimensional scenes should be tested on these models to ensure they work at 100 % potential. Research identifies the need to generate reliable models at a fast pace as the deepfake technology is rapidly evolving making the situation convoluted. The implementation of standardized protocols would help to train the deepfake video detection models for rigorous, complex real-world scenarios enhancing their resilience and adaptability [86].

9. Conclusion

The future of deepfake videos is poised to amplify technical challenges and ethical concerns. Advancements in machine learning algorithms, particularly in generative models like GANs, are enabling the creation of increasingly realistic deepfakes. As these algorithms become more accessible and refined, the barrier to generating sophisticated deepfake videos is lowering, raising the likelihood of misuse. Addressing the technical aspects of deepfake detection and attribution is crucial to developing effective countermeasures and safeguarding against the potential misuse of this technology for harmful purposes.

This research encloses a precise review of deepfake video detection procedures accentuating the potential strengths and flaws of feature-based, audio-based, and video-based detection models and multi-modal techniques. The reviews discuss about some strong detection models under controlled conditions for visual approaches like CNN (Convolutional Neural Networks) and frequency-domain. The research provides a clear picture regarding the challenges these advanced deepfake video detection models face when utilized for real-world cases where situations and factors are more complex than in controlled experiments. It talks about the ongoing challenges faced by detection models such as low-quality content, compression effects, and the fast-paced evolution of deepfake techniques that are too convoluted to catch. Similarly, audio-based methods, which excel at identifying inconsistencies in synthetic speech, are limited by the scarcity and quality of available reference datasets. In contrast, multi-modal approaches that integrate both audio and visual signals offer the most robust solutions, particularly in cases where manipulations in one modality are camouflaged by the other.

The accelerating development of deepfake generation technologies continues to outpace many detection techniques, highlighting the need for enhanced strategies. To stay ahead, future research must focus on improving the generalizability and resilience of detection systems in real-world contexts, where deepfakes exhibit diverse levels of quality, compression, and complexity. The advancement of multi-modal detection systems, alongside the creation of standardized benchmarks for model evaluation, should be prioritized. These measures are essential for equipping detection frameworks to counter the increasing sophistication of deepfakes, ensuring the integrity and security of digital media in a wide array of unpredictable environments.

CRediT authorship contribution statement

Mubarak Alrashoud: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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